What should an analyst look for in DNS, HTTP, HTTPS/TLS, ARP, ICMP and SMB traffic?

Table: protocol, normal behaviour, suspicious behaviour.

A SOC analyst monitors network traffic to identify normal communication, suspicious behaviour, and potential attacks.

The following protocols are especially important.

1. Protocol analysis table

Protocol	Normal behaviour	Suspicious behaviour
DNS	Clients resolve domain names to IP addresses.	DNS tunnelling, unusually long queries, excessive NXDOMAIN responses, suspicious domains, frequent TXT queries.
HTTP	Clients communicate with web servers using GET, POST and other methods.	Suspicious POST requests, malicious downloads, unusual user agents, directory traversal, command injection, abnormal HTTP status codes.
HTTPS/TLS	Encrypted communication between clients and servers.	Invalid certificates, unusual TLS versions, suspicious SNI, rare destinations, unusual certificate fingerprints, unexpected encrypted traffic patterns.
ARP	Devices resolve IPv4 addresses to MAC addresses on a local network.	ARP spoofing, duplicate IP addresses, unsolicited ARP replies, gateway MAC address changes.
ICMP	Ping, network diagnostics, and error reporting.	ICMP tunnelling, excessive echo requests, large unusual packets, suspicious ICMP sweeps.
SMB	Windows file sharing, printer sharing, and domain-related communication.	SMBv1 usage, suspicious administrative shares, brute-force attempts, unusual lateral movement, unexpected access to ADMIN$ or C$.

2. DNS (Domain Name System)



DNS translates domain names into IP addresses.

Example:

User enters:

www.google.com

DNS query:

www.google.com

DNS response:

IP address associated with the domain

Common DNS record types:

Record	Purpose
A	Maps a hostname to an IPv4 address
AAAA	Maps a hostname to an IPv6 address

CNAME	Alias for another hostname
MX	Mail server record
TXT	Text information, SPF, verification data
NS	Authoritative name server
PTR	Reverse DNS lookup
SOA	Zone authority information

Suspicious DNS indicators

I. Extremely long subdomain names.
II. High volumes of DNS queries from one endpoint.
III. Frequent requests to newly observed domains.
IV. Repeated NXDOMAIN responses.
V. Large numbers of TXT queries.
VI. Random-looking domain names.
VII. DNS requests to unusual external resolvers.

Attack example: DNS tunnelling

An attacker may encode stolen information inside DNS queries and send it to an attacker-controlled domain.

Example:

dGhpcy1pcy1zdG9sZW4tZGF0YQ.attacker.com

This resembles Base64-encoded content in a subdomain. One such query is not proof of tunnelling, but repeated encoded-looking queries deserve investigation.

Useful Wireshark filter:

dns

More specific:

dns.flags.response == 0

This shows DNS queries.

3. HTTP (Hypertext Transfer Protocol)

HTTP is an application-layer protocol used for web communication.

Common methods:

Method	Purpose
GET	Retrieve information
POST	Submit information
PUT	Upload or replace information
DELETE	Delete a resource
HEAD	Retrieve headers only



OPTIONS	Identify supported methods

Suspicious HTTP indicators

- POST requests to unusual URLs.
- Unexpected executable downloads.
- Suspicious user-agent strings.
- Repeated 401 or 403 responses.
- HTTP requests containing encoded commands.
- Directory traversal attempts such as ../.
- Requests to unusual ports.

Example suspicious request:

```
GET /../../etc/passwd HTTP/1.1
Host: example.com
```

This may indicate a directory traversal attempt.

Useful Wireshark filters:

```
http
http.request.method == "POST"
http.response.code >= 400
```

4. HTTPS/TLS

HTTPS is HTTP communication protected by TLS encryption.

TLS provides:

- Confidentiality.
- Integrity.
- Server authentication through certificates.
- Optional client authentication.



What should analysts inspect?

Even when application data is encrypted, analysts can often inspect:

- Source and destination IP addresses.
- Source and destination ports.
- TLS version.
- Server Name Indication (SNI), when visible.
- Certificate information.
- Certificate validity.
- Handshake failures.
- Connection duration and data volume.

Suspicious indicators

- Self-signed or invalid certificates where unexpected.
- Unusual or obsolete TLS versions.
- Connections to suspicious or newly observed domains.
- Unexpected certificates.
- Unusual outbound data volume.
- Repeated TLS handshake failures.

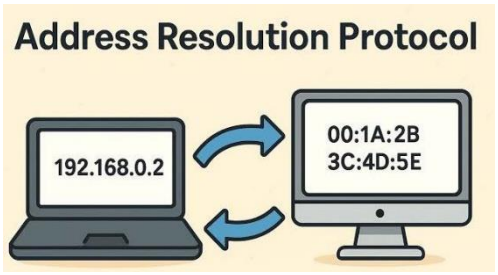
Useful Wireshark filters:

```
tls
tls.handshake.type == 1
```

The second filter identifies TLS ClientHello messages.

Important: TLS 1.3 encrypts more handshake information than earlier versions. With encrypted client hello (ECH), even the traditional SNI may not be visible. A SOC analyst should not assume that every TLS connection exposes the domain name.

5. ARP (Address Resolution Protocol)



ARP resolves an IPv4 address into a MAC address on a local network.

```
Example:
Who has 192.168.1.1?
192.168.1.1 is at AA:BB:CC:DD:EE:FF
```

Suspicious behaviour: ARP spoofing

An attacker sends forged ARP replies to associate their MAC address with another device's IP address, often the default gateway.

This can enable a Man-in-the-Middle (MITM) attack.

Indicators:

- Multiple MAC addresses claiming the same IP.
- Unexpected gateway MAC changes.
- Unsolicited ARP replies.
- Excessive ARP broadcasts.
- ARP replies without corresponding requests.

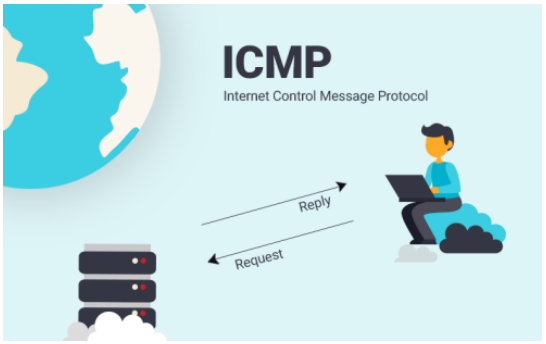
Wireshark filters:

```
arp
arp.opcode == 2
```

The second filter displays ARP replies.

6. ICMP (Internet Control Message Protocol)

ICMP supports network diagnostics and error reporting.



Common ICMP messages:

Type	Meaning
0	Echo Reply
3	Destination Unreachable
5	Redirect
8	Echo Request
11	Time Exceeded

Suspicious indicators

- ICMP sweeps across many IP addresses.
- Unusual ICMP payload sizes.
- High-volume ICMP traffic.
- ICMP traffic between systems that normally do not communicate.
- Possible ICMP tunnelling.

Example:

An attacker sends ICMP Echo Requests to a large range of internal addresses to discover live systems.

Wireshark filters:

icmp

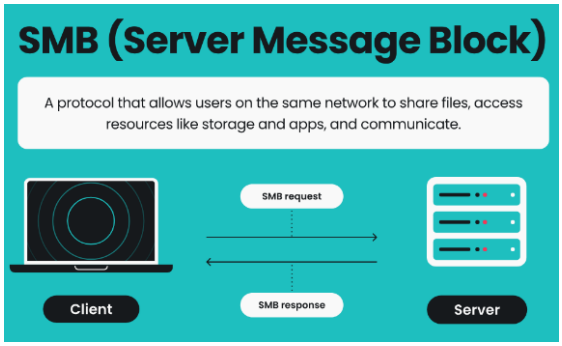
icmp.type == 8

icmp.type == 0

7. SMB (Server Message Block)

SMB is used primarily in Windows environments for:

- File sharing.
- Printer sharing.
- Remote administration.
- Named pipes.
- Access to shared resources.



Common SMB ports:

Port	Purpose
TCP 445	Direct-hosted SMB
TCP 139	NetBIOS session service (legacy SMB)

Suspicious SMB behaviour

- Unexpected access to administrative shares.
- SMB connections between ordinary workstations.

- Repeated authentication failures.
- SMBv1 usage in environments where it should be disabled.
- Unusual file transfers.
- Access to many systems from one workstation.
- Unexpected remote service or administrative activity.

Important attack examples

- Pass-the-Hash: An attacker uses a captured password hash to authenticate without knowing the plaintext password.
- Lateral movement: An attacker uses SMB to access another Windows machine.
- Ransomware propagation: Malware may use SMB shares to spread to other systems.

Useful filters:

```
smb
smb2
tcp.port == 445
```